

1 What the posterior identifies

Well specified, the posterior concentrates on the theories with the data's law and is the prior among them: it identifies the theorem set (under L1, parameters admissible) or the instance union (under L0), not the axioms. Splits of a schema are decided by Occam terms or by usage statistics; spare templates are never refuted and at best decay polynomially; stochastic positive data identify both ends of Gold's limit point. Misspecified, the posterior goes to the best-fitting generator, which can be weaker than the axioms behind the data, or unsound. Sources are the model track (§§2, 3, 5, 8) unless named; proofs and computations are in Appendix A. Logarithms are natural, KL in nats; the experiments report bits.

1.1 Well-specified consistency

Definition 1.1 (Setting W).

model §2 $\mathcal{H}$ is a countable class with a prior π , $\sum_T \pi(T) = 1$; each T carries a *fixed* law P_T on the countable set $\mathcal{S}$ (fixed weights). The data $D_n = (X_1, \dots, X_n)$ are i.i.d. from P_{T^*} , $\pi(T^*) > 0$. The posterior is $\pi_n(T) := \pi(T)P_T(D_n)/M(D_n)$, $M(D_n) := \sum_T \pi(T)P_T(D_n)$; the predictive law is $M_n := \sum_T \pi_n(T)P_T$; the *generator class* is $C^* := \{T : P_T = P_{T^*}\}$.

Theorem 1.2 (Concentration on the generator class). *[proved; known (Doob 1949)]*

model Thm 2.1; universal Lemma U7 Under W, $\pi_n(C^*) \rightarrow 1$ a.s.; moreover $\pi_n(T) \rightarrow \pi(T)1[T \in C^*]/\pi(C^*)$ for every T , and in total variation.

The proof is Doob's argument (Doob, 1949), written out for a countable class in Appendix A.1.

Corollary 1.3 (Inside C^* the posterior is the prior). *[proved; computed]*

model Cor 2.2 For $T, T' \in C^*$ and every n , $\pi_n(T)/\pi_n(T') = \pi(T)/\pi(T')$.

Corollary 1.4 (Limits of deductive beliefs). *[proved]^{model Cor 2.3}* Under W, a.s., for all s and $d \leq \infty$ at once, $\pi_n(\{T : s \in \text{Th}_d(T)\}) \rightarrow \pi(\{T \in C^* : s \in \text{Th}_d(T)\})/\pi(C^*)$.

The data decide a deductive question only through C^* ; inside it the prior decides. The limit is $1[s \in \text{Th}_d(T^*)]$ under L0 (full-support $\mathcal{Q}$, positive weights) for every d ($\text{supp } P_T = \bigcup \text{inst}(T)$) and under L1 with parameters admissible for $d = \infty$ ($\text{supp } P_T = \text{Th}(T)$; ???); under L2 only $\text{Th}_{\leq d}$ is fixed, and for Hänni's scores nothing of this kind holds.

Proposition 1.5 (Rates). *[proved]*

model Prop 2.4 Assume W. (a) For $T \notin C^*$, $\pi(T) > 0$: $\frac{1}{n} \ln \frac{\pi_n(T^*)}{\pi_n(T)} \rightarrow \text{KL}(P_{T^*} \| P_T) \in (0, \infty]$ a.s. (b) $\sum_{i=1}^n \mathbb{E} \text{KL}(P_{T^*} \| M_{i-1}) \leq \ln \frac{1}{\pi(C^*)}$. (c) For $A \subseteq \mathcal{H} \setminus C^*$ and $\rho_T := \sum_s (P_{T^*}(s)P_T(s))^{1/2}$, $\mathbb{E}[(\pi_n(A)/\pi_n(T^*))^{1/2}] \leq \sum_{T \in A} (\pi(T)/\pi(T^*))^{1/2} \rho_T^n$; so $P[\pi_n(A) \geq \varepsilon] \leq C \rho^n \varepsilon^{-1/2}$ if $\sup_A \rho_T \leq \rho < 1$ and $C := \sum_{T \in A} (\pi(T)/\pi(T^*))^{1/2} < \infty$ (true for π_λ , $\lambda \geq 2$).

The rate (c) degenerates for spare templates ($\rho_T \geq (1 - \varepsilon)^{1/2}$ for a spare of weight ε), whose decay the weight prior sets (Proposition 1.18).

1.2 Identification is of the generator

Proposition 1.6 (L1 separates deductively equivalent theories). *[proved (open parameter region); computed]*

model Prop 2.5 Let $a := \neg(S0 = 0)$, $b := \neg(SS0 = 0)$, $T_1 := \{a, b\}$, $T_2 := \{a, a \rightarrow b\}$ (ground, weights $\frac{1}{2}$); $\text{Th}(T_1) = \text{Th}(T_2)$. Under L1 or L2, $P_{T_1}(b) \geq \alpha_{\text{ax}}/2$ and $P_{T_2}(b) \leq \alpha_r/(1 - \alpha_r)$; so $P_{T_1} \neq P_{T_2}$ if $\alpha_{\text{ax}}/2 > \alpha_r/(1 - \alpha_r)$.

The laws also differ in every computed setting outside the proved region (Appendix A.2); for all parameters it is open. With data from P_{T_1} or P_{T_2} the posterior picks the generating axiomatisation, not a logically privileged one; from a third generator the drift is a difference of two KL divergences and can vanish.

Proposition 1.7 (Equal laws; splits are exact ties). *[proved; computed]*

model Prop 2.6 and remark; pa Prop 2.1

Let $\mathcal{Q}$ have full support and all weights be positive. (a) Under $\mathbf{L0}$, $P_{T,w} = P_{T',w'}$ implies $\bigcup \text{inst}(T) = \bigcup \text{inst}(T')$, hence $\text{Th}_d(T) = \text{Th}_d(T')$ for all d . (b) Let $\tau \in T$ have a metavariable M whose law factors at the root over finitely many productions, $\mathcal{Q}_M(\lambda \bar{z}.r(\beta_1, \dots, \beta_a)) = p_r \prod_j \mathcal{Q}(\beta_j)$, and let τ_r be τ with M 's body fixed to root r and fresh metavariables for its children. The root split T' , with τ replaced by the τ_r at weights $w'_{\tau_r} := w_\tau p_r$, consists of DT° templates, and $P_{T',w'} = P_{T,w}$ under $\mathbf{L0}$; $\mu_{T'} = \mu_T$, $Z_{T'} = Z_T$ under $\mathbf{L1}$, $\mathbf{L2}$. (c) A renamed or argument-permuted copy (AS Lemma A.2) has the same law if $\mathcal{Q}$ is symmetric in holes. Equal laws give equal instance sets, hence equivalence for first-order templates with only 0-ary term metavariables (AS Prop C.1); in general (e.g. A1, or DT°) this converse is open.

So with weights matched to $\mathcal{Q}$ no data tell a schema (e.g. T_{Ind}) from its root split; only the prior does. The tie needs $\mathcal{Q}$ to factor at the root, which track pa's learned grammar does not (??).

Remark 1.8 (Chain likelihood, latent citations). *[proved; computed]*^{experiments Prop X13, E8, E6} In the chain Ch_J the tie holds for a 0-ary closed-guard term metavariable of a component that is not an MP major premise; in E8, where a datum $\psi(t)$ may be cited or derived from $\varphi(t)$ by MP, the two log-probabilities agree to $1.1 \cdot 10^{-13}$. In E6 the chain $\mathbf{L1}$ identifies the generating form of commutativity by $n = 32$, among logically equivalent and among instance-equivalent forms; under $\mathbf{L1}_{\text{cit}}^{\text{sel}}$ five closed-guard forms tie exactly (??).

1.3 Positive data and Gold

Theorem 1.9 (Identification in the limit with probability one). *[(a), (b) proved; (b') known, proof sketch; (c) conjecture]*

model Thm 5.1

Let $\mathcal{H}$ be all theories, with a prior positive on T^* . (a) For fixed generators ($\mathbf{L0}$ with fixed positive weights; or $\mathbf{L1}$ with fixed positive parameters, $m \leq 1$, parameters admissible; full-support $\mathcal{Q}$) and data i.i.d. from P_{T^*} , the learner $g_n := L$ if $\pi_n(\{T : \text{supp } P_T = L\}) > \frac{1}{2}$ equals $\text{supp } P_{T^*}$ for all large n , a.s.: $\bigcup \text{inst}(T^*)$ under $\mathbf{L0}$, $\text{Th}(T^*)$ under $\mathbf{L1}$. No bound on the number of templates is needed. (b) With Dirichlet weights (prior $\pi(T)\text{Dir}_T(dw; \alpha)$, $\mathbf{L0}$, full-support $\mathcal{Q}$, data i.i.d. from P_{T^*,w^*}), for Lebesgue-a.e. w^* , $\pi_n(\{T : \bigcup \text{inst}(T) = \bigcup \text{inst}(T^*)\}) \rightarrow 1$ a.s. (b') Under (b), $\pi_n(\{(T, w) : \text{TV}(P_{T,w}, P_{T^*,w^*}) < \varepsilon\}) \rightarrow 1$ a.s. for all $\varepsilon > 0$ (Ghosal and van der Vaart, 2017, Thm 6.9; numbering checked only via citing papers). (c) Conjecture: (b) holds for every w^* in the open simplex.

(b) runs Doob's argument with the support functional $\bigcup \text{inst}(T)$, which takes countably many values and a.s. equals the set of sentences seen (Appendix A.3).

Remark 1.10 (The exact class has posterior mass zero). *[refuted; counter-statement proved]*

model Thm 5.1(b'')

The pre-referee (b) read “ $\pi_n(\{(T, w) : P_{T,w} = P_{T^*,w^*}\}) \rightarrow 1$ for a.e. w^* ”. If $|T^*| \geq 2$ with linearly independent component laws, this mass is 0 at every n for all w^* off a countable set: each $\{w : P_{T,w} = P_{T^*,w^*}\}$ is Lebesgue-null, and the posterior on w is absolutely continuous with respect to Dir_T .

Remark 1.11 (Which computed posteriors are covered). *[known]*

experiments Prop X5

Theorem 1.2 covers the experiments' Dirichlet posteriors only for one-component

generators (E1, E4, E6). E2, E3(b), E5(b), E8 sample at one fixed weight vector, where only (b) and conjecture (c) speak: a yardstick, not an instance of a theorem.

Proposition 1.12 (L_∞ against L_5). *[proved]^{model Prop 5.2} Let $L_1 \subsetneq L_2 \subsetneq \dots$, $L_\infty := \bigcup_i L_i$, each L_i ($i \leq \infty$) with a generator P_i , $\text{supp } P_i = L_i$, whose class has prior mass $\pi_i > 0$ (setting W). (a) On data i.i.d. from P_i , $g_n = L_i$ eventually, a.s. (b) Some text for L_∞ has $g_n \neq L_\infty$ infinitely often (Gold, 1967). (c) Such texts have P_∞^∞ -probability 0.*

The learner has no fixed preference between the ends of the limit point; the size principle decides (on P_5 -data the ratio of L_∞ 's generator to L_5 's is $P_\infty(L_5)^n$ times a factor of mean at most 1; ??). E5 shows both ends identified and the MAP changing at each of 14 stages of Gold's text (??). For the IS_n chain both directions separate, after presumably astronomical waiting times (a heuristic; Appendix A.3, ??).

Remark 1.13 (Memorisation and the bound). *[proved; known (prior work, flagged)]*

^{model §5.2} A fixed memoriser $\text{Mem}(E)$ dies once a datum outside E appears; before that it pays its prior and, with Dirichlet weights, about $\frac{|E|-1}{2} \ln n$ nats. The memoriser *class* can keep mass $\exp(-O(\ln^2 n))$ (??). So the Bayesian needs no bound on the number of templates, unlike the cautious learner (AS Prop 6.4). Identification from stochastic positive data goes back to Horning (1969) and Angluin (1988) (existence checked, content not).

1.4 Splits and the MDL finding

AS §6.10 found that MDL “tracks the statistics of usage, not the logical boundaries of schemas” (a naive code splits T_{Ind} by a linear margin, a well-specified one gains at most $O(\log n)$; AS Props E.16, E.17). Does a derivation likelihood change this? With learned weights a root split must learn the root law that the schema takes from $\mathcal{Q}$.

Proposition 1.14 (Schema against root split, L0). *[proved; known (the χ^2 limit); computed]*

^{model Prop 5.4} Let $T = \{\tau\}$, T' its root split over K productions with $\text{Dir}(\alpha)$ weights, the data n instances of τ with root law r , root counts n_f , $\hat{r} := n_f/n$, and p the root law of $\mathcal{Q}$. (a) $\Delta_n := \ln P_{T'}^{\text{Dir}}(D_n) - \ln P_T(D_n) = \ln \text{DirMult}(n_f; \alpha) - \sum_f n_f \ln p_f$, $\text{DirMult}(n_f; \alpha) := \frac{\Gamma(K\alpha)}{\Gamma(K\alpha+n)} \prod_f \frac{\Gamma(\alpha+n_f)}{\Gamma(\alpha)}$. (b) If all $n_f \rightarrow \infty$: $\Delta_n = n\text{KL}(\hat{r}||p) - \frac{K-1}{2} \ln n + c_{K,\alpha} + (\alpha - \frac{1}{2}) \sum_f \ln \hat{r}_f + o(1)$, $c_{K,\alpha} := \ln \frac{\Gamma(K\alpha)}{\Gamma(\alpha)^K} + \frac{K-1}{2} \ln 2\pi$. (c) If $r = p$: $\Delta_n = -\frac{K-1}{2} \ln n + O_P(1)$ ($2n\text{KL}(\hat{r}||p) \rightarrow \chi_{K-1}^2$); the schema wins at rate $n^{-(K-1)/2}$. (d) If $r \neq p$: $\Delta_n/n \rightarrow \text{KL}(r||p) > 0$ a.s.; the split wins linearly. (e) The prior adds $\lambda(\ell(T') - \ell(T)) > 0$ bits for the schema.

Proposition 1.15 (Schema against root split, derivation likelihoods). *[(a)–(c) proved; (d) conjecture; computed]^{model Prop 5.6} Let $T = \{\tau\}$ root-factor as in Proposition 1.7(b), $|R| = K$, root law p interior, T' the split with $\text{Dir}(\alpha)$ weights, under L0, L1 ($m < 1$) or L2 with other parameters fixed and shared; $P^{(p')}$ is the law of $\{\tau\}$ with root law p' , and $\text{BF}_n := P_{T'}^{\text{Dir}}(D_n)/P_T(D_n)$. (a)*

$P_{T',w'} = P^{(w')}$, so $P_{T'}^{\text{Dir}}(D) = \int \prod_i P^{(p')}(X_i) \text{Dir}(dp')$: the split with learned weights is the schema with a learned root law. (b) If $\mathcal{Q}$ fits the usage (data i.i.d. from P_T), BF_n is a nonnegative supermartingale from 1: $P(\sup_n \text{BF}_n \geq 1/\delta') \leq \delta'$. (c) If not (data i.i.d. from $P^{(r)} \neq P^{(p)}$, r interior), $\frac{1}{n} \ln \text{BF}_n \rightarrow \text{KL}(P^{(r)}||P^{(p)}) \in (0, \infty]$ a.s. (d) Conjecture: under (b), with $p' \mapsto P^{(p')}$ identifiable and nonsingular Fisher information, $\ln \text{BF}_n = -\frac{K-1}{2} \ln n + O_P(1)$.

(b) is Ville's inequality (IL Lemma C.4); (c) uses a continuity lemma (Appendix A.4); (d) would need a Laplace theorem for a model with latent citations. Computed: Tables 2 and 3 (slope -0.507 for (d)).

Remark 1.16 (E8: latent citations). [computed]

experiments E8 For $\{\varphi(z), \psi(z), \varphi(z) \rightarrow \psi(z)\}$ against the split of $\varphi(z)$ by its term's root ($K = 4$; chain L1 data, 10 seeds), well-specified usage makes the split lose 1.43 bits per doubling of n under L1 and 1.51 under L0 ($n = 256 \dots 4096$; conjectured 1.5), and skewed usage makes it win 0.202 against 0.074 bits per datum ($n = 1024 \dots 4096$): a ψ -datum may be φ followed by MP, so under L1 its term informs the split too (??).

Remark 1.17 (What this says about the MDL finding). [proved; computed (as cited)]

model §5.3; pa §2; universal §10; experiments E8 As far as proved, a derivation likelihood does not change the MDL finding: with usage that $\mathcal{Q}$ fits, the split loses under L0 and gains at most $\ln(1/\delta')$ nats with probability $1 - \delta'$ under L1 (that it loses there is conjecture (d)); with usage that $\mathcal{Q}$ misfits, it wins linearly under both. Three qualifications. (i) Partial splits that only L1 keeps alive are not covered: a split of T_{Ind} with a non-atomic connective derives every induction instance, an unseen-connective datum costing 369–489 bits in pa's code instead of being impossible (??); a general comparison is open. (ii) A learned grammar can reverse the Occam drift: under pa's shared positional grammar the split into roots F costs $\Delta_{\text{prior}} + \frac{|F|-9}{2} \log_2 n + O(1)$ bits more than the schema, whose grammar pays for root symbols it never uses (??); with seven roots the split gains 1 bit per doubling yet is 762 bits behind at $n = 256000$. (iii) Unmodelled selection of the reported data creates linear split wins (??). In short: when the instantiation model is misspecified, the posterior tracks usage, not the logical boundary of a schema.

1.5 Spare templates

Proposition 1.18 (Cost of a spare template). [(a), (d1) proved; (b), (c), (d2) proof sketch; computed] model Prop 5.5; experiments Prop X7; pa Prop 5.1 Let the truth be (T^*, w^*) under L0 with $\text{Dir}(\alpha_\tau)$ priors, $A := \sum_{\tau \in T^*} \alpha_\tau$, σ a spare with parameter α_σ , $R_n := P_{T^* \cup \{\sigma\}}^{\text{Dir}}(D_n) / P_{T^*}^{\text{Dir}}(D_n)$. (a) If no datum is in $\text{inst}(\sigma)$, exactly $R_n = \frac{\Gamma(A+\alpha_\sigma)\Gamma(A+n)}{\Gamma(A)\Gamma(A+\alpha_\sigma+n)} \sim \frac{\Gamma(A+\alpha_\sigma)}{\Gamma(A)} n^{-\alpha_\sigma}$; at $\alpha_\sigma = \frac{1}{2}$, $\frac{1}{2} \log_2 n$ bits plus a constant and the prior bits. (b) If σ covers data but $\mathcal{Q}_\sigma(\text{supp } P^*) = 1 - c < 1$: $\ln R_n = -\alpha_\sigma \ln n + O_P(1)$. (c) If $\text{inst}(\sigma) \subseteq \text{supp } P^*$, $\mathcal{Q}_\sigma$ outside the span of T^* 's components: $\ln R_n = -\frac{\alpha_\sigma}{2} \ln n + O_P(1)$. (d1) If $\mathcal{Q}_\sigma = \mathcal{Q}_{\tau_1}$, $\tau_1 \in T^*$ (e.g. an argument-permuted copy), and T^* 's instance sets are disjoint, $R_n \rightarrow \frac{\Gamma(A+\alpha_\sigma)\Gamma(\alpha_1)}{\Gamma(A)\Gamma(\alpha_1+\alpha_\sigma)} (w_1^*)^{\alpha_\sigma} > 0$ a.s.: the spare never vanishes. (d2) If $\mathcal{Q}_\sigma$ is in the convex hull of T^* 's components but equals none, and the induced prior on $P_{T^* \cup \{\sigma\}, w}$ has a continuous positive density at P^* , R_n tends to a positive constant.

(b) and (c) need a boundary Laplace approximation (not written out) of the kind studied for overfitted mixtures by Rousseau and Mengersen (2011). In E5 unused spares follow (a) exactly and a nested spare, case (c), has fitted exponent -0.251 ± 0.002 on $n = 10^2 \dots 10^8$; without the density hypothesis of (d2) a spare can gain slowly (Table 4).

Remark 1.19 (All spares; what spares cost). [proved (the sum); computed] model §5.4; experiments E2, E5 With a common $\alpha_\sigma = \alpha$, R_n in (a) does not depend on σ ; so the posterior mass of all $T^* \cup \{\sigma\}$ with σ covering no datum is at most $R_n / \pi(T^*) = O(n^{-\alpha})$. With fixed weights the factor is $(1 - w_\sigma)^n$. Spares inside the support do not threaten identification of the instance union. What spares cost is slow rejection: about $(\text{prior ratio} / \delta_r)^{1/\alpha}$ data to fall below δ_r in case (b). A spare sentence that T^* refutes makes $T^* \cup \{\sigma\}$ inconsistent, and positive data remove it only like $n^{-1/2}$ (E2, 0 = S_0 beside Q1: posterior 0.004 at $n = 512$); the size principle does not see inconsistency (??).

1.6 Misspecification

Human data are theorems of the actual axioms, but are not drawn from any P_T of the model.

Theorem 1.20 (Finite class: concentration on the KL-minimisers). *[proved; known (general form)]*
model Thm 3.1 Let $\mathcal{H}$ be finite, $\pi > 0$, the data i.i.d. from any P_H , $K_T := \text{KL}(P_H \| P_T)$, $\min_T K_T < \infty$, $\mathbb{M} := \arg \min_T K_T$. Then $\pi_n(\mathbb{M}) \rightarrow 1$ a.s., with $\frac{1}{n} \ln \frac{\pi_n(T)}{\pi_n(T_0)} \rightarrow K_{T_0} - K_T < 0$ for $T \notin \mathbb{M} \ni T_0$; inside $\mathbb{M}$ the posterior need not converge.

The general form is known (Berk, 1966; Kleijn and van der Vaart, 2006) (abstracts read, conditions not); it need not describe the full template class (Remark 1.24).

Example 1.21 (The axiom's own theorems lead to a weaker theory). *[proved]*^{model Ex 3.2} In L_A with parameters not admissible, under **L0** with fixed full-support $\mathcal{Q}$ and $\pi(\{0+z=z\}) > 0$, let the human's axiom be $\forall x(0+x=x)$ and the data its theorems $0+t=t$, $t \sim \mathcal{Q}$. They are well specified for $\{0+z=z\}$, whose generator class has instance union $I_c(0+x=x)$ (Proposition 1.7), and no such theory proves $\forall x(0+x=x)$ (take $\mathbb{N} \cup \{a\}$ with $0+a:=0$). So $P(T \vdash \forall x(0+x=x) \mid D_n) \rightarrow 0$ a.s.: the ω -gap (AS §4.5), forced by the likelihood.

Under **L1** without parameters $\text{supp } P_T = \text{Th}(T)$ fails; ?? treats that case.

Lemma 1.22 (Sound templates cover at most one zero-sum equation). *[proved]*^{model Lemma 3.3} Let $Z := \{t = 0 : t \text{ a closed term over } 0, +\}$. A DT° template all of whose instances are true in $\mathbb{N}$ (parameters, if admissible, read universally) covers at most one member of Z .

Example 1.23 (An unsound limit). *[proved; computed]*^{model Ex 3.4} Let the human's axioms be $\mathcal{Q}$ and the data true equations $t = 0$, $t \in Z$, from a law of infinite support; the class is finite, made of sound theories $T_1, \dots, T_m$ and $T_{\text{esc}} := \{z = 0\}$, all with positive prior; **L0** with full-support $\mathcal{Q}$. By Lemma 1.22 the T_j cover finitely many members of Z , so a.s. $\pi_n(T_{\text{esc}}) = 1$ eventually. But $T_{\text{esc}} \vdash S0 = 0$ and $\mathcal{Q} \vdash \neg S0 = 0$; $\mathcal{Q}$ itself, if in the class, dies at the first datum, as its axioms are not in Z . Computed: against twenty best-possible sound theories at prior 0.99, $S0 = 0$ reached posterior ≥ 0.99 in 2000/2000 runs (median 17 data).

Remark 1.24 (The full class). *[conjecture; computed]*

model Rem 3.5 In the full class with Dirichlet weights the KL-projection picture can fail: theories with finite KL are unsound (Lemma 1.22), with infimum KL 0 approached by hybrids of memorisation and T_{esc} ; memorisers are sound but each has infinite KL, yet as a family they code the data with $o(n)$ extra bits. Conjecture: unsound mass $\rightarrow 1$. Computed for three hand-picked families: at $n = 10^4$ the hybrid leads memorisation by at least 7000–25000 bits (Table 5).

In Example 1.23 $\mathcal{Q}$ fails because the data are not axiom instances; a derivation likelihood removes that failure, but not reliably its outcome.

Example 1.25 (A derivation likelihood does not reliably rescue the axioms). *[computed]*
model Ex 3.6 **L1^{eq}** derives closed equations over $0, S, +$ (cite, refl, sym, trans, congruences), terms of size $\leq N$, $P_T = \mu_T/Z_T$ exact. Class: $T_{\mathcal{Q}} := \{z + 0 = z, z_1 + Sz_2 = S(z_1 + z_2)\}$ (Q4, Q5) and $T_{\mathcal{Q}} \cup \{0 + z = z\}$ (sound); T_{esc} and $T_{\mathcal{Q}} \cup \{z = 0\}$ (unsound). On true equations $t = 0$ over $\{0, +\}$ the cross-entropy minimiser is unsound in all 16 settings in which $\mathcal{Q}$ gives $+$ probability 0.25, and sound in 5 of 8 when $+$ is rare under $\mathcal{Q}$ (Table 6): in grammar 1, citing $z = 0$ pays about 2 nats per $+$ through $\mathcal{Q}$, deriving from Q4, Q5 about 7. Limits: truncated, equational only, three grammars, a finite class.

Conjecture 1.26. *[conjecture]*

model Conj 3.7 In untruncated **L1^{eq}**, $\ln P_{T_{\mathcal{Q}}}(t = 0) = -c_{\text{der}}\#_+(t) + O(1)$ on balanced $\{0, +\}$ -terms; so T_{esc} beats $T_{\mathcal{Q}}$ on laws with many $+$ iff $\ln \frac{1}{p_+p_0} < c_{\text{der}}$, p_+, p_0 the root probabilities under $\mathcal{Q}$ (grammar 1: ≈ 2 against ≈ 7).

data (true unless stated)	likelihood	posterior lands on	where
<i>same theorems or instances, other axioms</i>			
instances, usage that $\mathcal{Q}$ misfits	L0, L1	root splits; nested C_2	Propositions 1.14 and 1.15 and Remark 1.27
PA instances, connective-rich motives	L0	$T^* + T_{\exists}$; observed-root fragments	??
<i>deductively weaker</i>			
closed instances of $\forall x(0 + x = x)$	L0	closed schema $0 + z = z$	Example 1.21
instances: numerals only; finite set	L0, L1	R_m , m growing; $\text{Mem}(D_n)$	Remark 1.27
PA instances, atomic motives	L0	$\mathbf{Q} + \{T_-, T_<\}$	????
narrow induction practice	pa's code	$\mathbf{Q} + T_E + T_N \subseteq \text{IS}_1$ (extrapolated)	??
theorems, no direct induction uses	L1^{sch}	$\mathbf{Q} + \text{a finite library}$	??
<i>unsound</i>			
derived equations $t = 0$, finite class	L0	$T_{\text{esc}} = \{z = 0\}$	Example 1.23
the same	L1^{eq}	T_{esc} or $T_{\mathbf{Q}} \cup \{z = 0\}$ (16/16 at $p_+ = 0.25$)	Example 1.25
the same, full class	L0, Dir.	memorisation- T_{esc} hybrids (conjecture)	Remark 1.24
prime values of $k^2 + k + 41$	L1^{sch}	the false schema (counterexample absent)	??
10% false near misses (not true)	L0, L1	verifier accepts falsehoods, 100/100	??
small n , rarely cited axioms	L0; L1^{sch}	unsound lumps (E2: 5/25 seeds)	????

Table 1: Kinds of misspecification and where the posterior lands; statuses are those of the cited items. Computed results hold within their pools or candidate sets.

Remark 1.27 (E3(a): misspecified usage of instances). [computed]

experiments E3(a) On instances of $x + 0 = x$ with a term law not the model's, the posterior moves at a linear rate to the best-fitting member of a family: nested $\{\varphi(z), \varphi(Sz), \varphi(SSz)\}$ (heavy tails; same instances), closed-guard splits R_m with m growing in n (numerals; $m = 9\text{--}10$ at $n = 512$, $11\text{--}12$ at 1024 ; $\varphi(2+3)$ underived), $\text{Mem}(D_n)$ (finite selected support). No unsound mass at large n ; completeness is lost (??).

Table 1 collects the cases: misspecified usage of axiom instances gives the same or weaker theorems; derived theorems, absent counterexamples, mistakes and small samples can give unsound ones.

1.7 Predictive versus deductive

Theorem 1.28 (Predictive versus deductive). [proved]

model Thm 8.1 Assume W . (a) $\sum_{n \geq 0} \mathbb{E} \text{KL}(P_{T^*} \| M_n) \leq \ln \frac{1}{\pi(C^*)}$. (b) $\pi_n(\{T : T \vdash s\}) \rightarrow \pi(\{T \in C^* : T \vdash s\}) / \pi(C^*)$ a.s., which is $1[T^* \vdash s]$ iff the members of C^* agree on s up to prior-null sets; they do when the generator determines the theorem set (L0 with full support; L1 with parameters admissible), and need not under L2 or when $\text{supp } P_T \subsetneq \text{Th}(T)$. (c) If $T_1 \vdash s$, $T_2 \not\vdash s$ and the data come from P_{T_1} , the expected log-odds of T_1 against T_2 grow by exactly $\text{KL}(P_{T_1} \| P_{T_2})$ per datum, while the per-datum predictions differ in total variation by at most $(\text{KL}(P_{T_1} \| P_{T_2})/2)^{1/2}$.

So when two theories differ in what they prove but hardly in what they generate, prediction is settled early and the deductive question stays near its prior odds for about $1/\text{KL}$ data. Misspeci-

fied, predictions go to the KL-projection and beliefs to its theorems, so predictive success certifies nothing deductive. For $\forall x\varphi$ this is the gap between ?? and ?? (compare Hutter, 2007).

Proposition 1.29 (Data that come with their proofs). *[proved]*

model Prop 8.2 Let each datum be a whole valid L1 tree from $P_T(\cdot \mid \text{valid})$, showing each node's rule and conclusion and whether a citation leaf cites a theory or a logical axiom, but not which template; weights fixed. (a) $P_T(\pi) = \prod_v P_T^{\text{L0}}(\text{concl}(v)) G(\pi)/Z_T$ over the theory-citation leaves v , with $P_T^{\text{L0}} := \sum_\tau w_\tau Q_\tau$, G independent of T , and Z_T a function of P_T^{L0} . (b) $P_T = P_{T'}$ on trees iff $P_T^{\text{L0}} = P_{T'}^{\text{L0}}$: the posterior concentrates on the L0 generator class of T^* , with instance union $\bigcup \text{inst}(T^*)$, and splits still tie. (c) So proofs identify $\bigcup \text{inst}(T^*)$, finer than the $\text{Th}(T^*)$ that conclusions identify (Proposition 1.6).

Remark 1.30 (Near misses). *[proved (the transfer)]*

model Rem 8.3 Hänni's "grade near misses" can be a corruption channel $K(s' \mid s)$ favouring s' near s , $P_T^K(s') := \sum_s P_T(s) K(s' \mid s)$. Each P_T^K is a fixed law with a data-independent normaliser, so Theorem 1.2, Corollary 1.4, ?? (with C_d^* defined through P^K) and the size principle hold when the channel is part of the well-specified model. The generator class can grow, since K need not be injective; by how much, for natural metrics, is open. The noise mixtures of ?? are related.

A Proofs and computations for Section 1

Proofs follow the model track's final notes; steps they leave out are named. The checks are the seeded scripts of `research/tracks/model/checks/` (c2-c16), each with its output file; the referee's independent scripts (r1-r7) are in `referee_code/`. The model track's referee re-did the proofs of Theorem 1.2, Corollaries 1.3 and 1.4, Proposition 1.5, Proposition 1.6, Proposition 1.7(a), (b), Theorem 1.20, Examples 1.21 and 1.23 and Lemma 1.22, Theorem 1.9(a), Proposition 1.12, Proposition 1.14, Proposition 1.18(a) and Theorem 1.28 line by line and found no error beyond the corrections recorded in Remark 1.10 and ??. The results added after the referee report (Theorem 1.9(b) with its new proof, Proposition 1.15, Proposition 1.18(d), Example 1.25, Proposition 1.29, Remark 1.30) were checked by the track and again while writing this appendix, not by a referee. Nothing is checked in a proof assistant.

A.1 Consistency and rates

Proof of Theorem 1.2. *Step 1 (joint law; Lévy).* Let Π be the law of $(T, X_1, X_2, \dots)$ on $\mathcal{H} \times \mathcal{S}^\infty$ with $\Pi(T = t, X \in E) = \pi(t) P_t^\infty(E)$, and $M := \sum_t \pi(t) P_t^\infty$ its data marginal. Since $\mathcal{H}$ is countable, Bayes' rule gives $\pi_n(A) = \Pi(T \in A \mid X_1, \dots, X_n)$ on the M -full event where $M(D_n) > 0$. By Lévy's upward theorem (known), for each fixed $A \subseteq \mathcal{H}$, $\pi_n(A) \rightarrow \Pi(T \in A \mid X_\infty)$ Π -a.s.; the convergence event is a data event, so it holds M -a.s. *Step 2 (the data determine the generator).* For $x \in \mathcal{S}^\infty$ let $F(x)(s) := \lim_n n^{-1} \#\{i \leq n : x_i = s\}$, and $F(x) := \perp$ if some limit fails to exist or the limits do not sum to 1. By the strong law applied to the countably many s , $P_t^\infty(F = P_t) = 1$ for each t (on that event the limits are the numbers $P_t(s)$, which sum to 1). Hence $\Pi(P_T = F(X)) = 1$. *Step 3 (the limit lives on $\{T : P_T = F(X)\}$).* Choose a version of $\{\Pi(T = t \mid X_\infty)\}_t$ that sums to 1 a.s. Each indicator $1\{P_t = F(X)\}$ is X_∞ -measurable, so $\mathbb{E}[1\{P_T = F(X)\} \mid X_\infty] = \sum_t 1\{P_t = F(X)\} \Pi(T = t \mid X_\infty)$ by monotone convergence. The left side is 1 a.s. by Step 2. $F(X)$ ranges Π -a.s. over the countable set $\{P_t : t \in \mathcal{H}\}$; apply Step 1 to the countably many sets $A_\nu := \{t : P_t = \nu\}$ and intersect the full-measure events. Then, M -a.s., $\lim_n \pi_n(A_{F(X)}) = \Pi(T \in A_{F(X)} \mid X_\infty) = 1$. *Step 4 (transfer to the truth).* $M \geq \pi(T^*) P_{T^*}^\infty$, so

every M -null event is $P_{T^*}^\infty$ -null and Step 3 holds $P_{T^*}^\infty$ -a.s. Under $P_{T^*}^\infty$, $F(X) = P_{T^*}$ a.s. by Step 2, so $A_{F(X)} = C^*$, and $\pi_n(C^*) \rightarrow 1$. *Pointwise and total variation.* Write $q_T := \pi(T)1[T \in C^*]/\pi(C^*)$. By Corollary 1.3, $\pi_n(T) = \pi_n(C^*)q_T$ for $T \in C^*$, so

$$\sum_T |\pi_n(T) - q_T| = \sum_{T \in C^*} q_T (1 - \pi_n(C^*)) + \pi_n(\mathcal{H} \setminus C^*) = 2(1 - \pi_n(C^*)) \rightarrow 0.$$

□

The universal track proves the same statement independently (its Lemma U7, total variation via Scheffé's lemma); the exact identity above is ours and gives the same conclusion directly.

Proof of Corollary 1.3. Every $T \in C^*$ has $P_T = P_{T^*}$ as a law on $\mathcal{S}$, so $P_T(D_n) = P_{T^*}(D_n)$ for every sequence, and the posterior ratio is the prior ratio. □

Computed (c9b; every likelihood by matching; three seeds, $n \leq 2000$): $\{z+0=z\}$ and its root split with weights $p_f = (0.5, 0.3, 0.1, 0.1)$ agree to $5.68 \cdot 10^{-14}$ in log-likelihood; an over-general template, a split missing a root and T^* plus the spare $0 = S0$ (weight 0.05, decaying as 0.95^n) go to 0. An earlier check (c9) was tautological for the split (referee m5).

Proof of Corollary 1.4. Put $A := \{T : s \in \text{Th}_d(T)\}$. Then $\pi_n(A) = \pi_n(A \cap C^*) + \pi_n(A \setminus C^*)$. The second term is at most $\pi_n(\mathcal{H} \setminus C^*) \rightarrow 0$. By Corollary 1.3 the first equals $\pi_n(C^*)\pi(A \cap C^*)/\pi(C^*)$. Both hold on the single event of Theorem 1.2, so simultaneously for all (s, d) . For the two cases named after the corollary: under L0 with full-support $\mathcal{Q}$ and positive weights, $\text{supp } P_T = \bigcup \text{inst}(T)$, so every $T \in C^*$ has T^* 's instance union and hence its Th_d ; under L1 with parameters admissible, $\text{supp } P_T = \text{Th}(T)$, so every $T \in C^*$ has $\text{Th}(T) = \text{Th}(T^*)$. □

Proof of Proposition 1.5. (a) $\ln \frac{\pi_n(T^*)}{\pi_n(T)} = \ln \frac{\pi(T^*)}{\pi(T)} + \sum_{i \leq n} Y_i$ with $Y_i := \ln \frac{P_{T^*}(X_i)}{P_T(X_i)}$ i.i.d. Their negative part is integrable, since $(\ln x)^+ \leq x$ gives $\mathbb{E}Y^- = \mathbb{E}(\ln \frac{P_T}{P_{T^*}})^+ \leq \mathbb{E} \frac{P_T(X)}{P_{T^*}(X)} \leq 1$, and $\mathbb{E}Y = \text{KL}(P_{T^*} \| P_T) > 0$ by Gibbs' inequality, as $P_T \neq P_{T^*}$. The strong law holds with limit in $(0, \infty]$. If $P_T(X_i) = 0$ for some i , the ratio is $+\infty$ from then on. (b) $M(D_n) \geq \pi(C^*)P_{T^*}(D_n)$. By the chain rule $\ln \frac{P_{T^*}(D_n)}{M(D_n)} = \sum_{i \leq n} \ln \frac{P_{T^*}(X_i)}{M_{i-1}(X_i)}$, and the conditional expectation of the i -th term given D_{i-1} is $\text{KL}(P_{T^*} \| M_{i-1})$. Take expectations. (c) $\frac{\pi_n(A)}{\pi_n(T^*)} = \sum_{T \in A} \frac{\pi(T)}{\pi(T^*)} L_T$ with $L_T := \prod_{i \leq n} \frac{P_T(X_i)}{P_{T^*}(X_i)}$. Use $(\sum a_T)^{1/2} \leq \sum a_T^{1/2}$ and, by independence, $\mathbb{E}L_T^{1/2} = \rho_T^n$. For the probability bound, $\pi_n(A) \leq \pi_n(A)/\pi_n(T^*)$ and Markov's inequality applied to the square root. □

For the degeneration: if $T = T^* \cup \{\tau\}$ with weight ε on τ and T^* 's weights scaled by $1 - \varepsilon$, then $P_T \geq (1 - \varepsilon)P_{T^*}$, so $\rho_T \geq (1 - \varepsilon)^{1/2} \sum_s P_{T^*}(s) = (1 - \varepsilon)^{1/2}$.

A.2 Identification of the generator

Proof of Proposition 1.6. $\text{Th}(T_1) = \text{Th}(T_2)$: b follows from a and $a \rightarrow b$ by MP, and $a \rightarrow b$ follows from b by an A1 instance and MP. Under T_2 : b has root $\neg$; every instance of a logical axiom has root $\rightarrow$, $\forall$ or $=$, and b is not an axiom of T_2 ; so no one-node tree concludes b , and $\mu_{T_2}(b) \leq \text{Pr}[\text{the root is a rule node}] = \alpha_r$. With $Z_{T_2} \geq 1 - \alpha_r$ (??(b)), $P_{T_2}(b) \leq \alpha_r/(1 - \alpha_r)$. Under T_1 the one-node tree citing b has probability $\alpha_{\text{ax}}/2$, and $Z_{T_1} \leq 1$. The same argument holds under L2. □

Computed (c6; A1–A3 and MP, formulas of size ≤ 3 , depth ≤ 2 ; rule probabilities for (ax, lg, MP) (0.6, 0.25, 0.15) in the proved region, (0.5, 0.3, 0.2) on its boundary, (0.3, 0.3, 0.4) and (0.2, 0.4, 0.4) outside): $P_{T_1}(b) = 0.16\text{--}0.35$, $P_{T_2}(b) = 0.007\text{--}0.04$, $\text{TV} = 0.16\text{--}0.35$; the referee's

implementation (r6) confirms both bounds. A route to all parameters, not carried out: $P_T(s)$ is a power series in the rule probabilities with nonnegative coefficients, so the difference is real-analytic on the subcritical region and nonzero near $\alpha_r = 0$.

Proof of Proposition 1.7. (a) $\text{supp } P_{T,w} = \bigcup \text{inst}(T)$ by ??(c) with positive weights, and Th_d depends only on the instance union (??). (b) *Templates.* Substituting the body into a pattern occurrence $M(\bar{y})$ gives $r(N_1(\bar{y}, \dots), \dots)$; each N_j has a pattern occurrence, because $\bar{y}$ are distinct bound variables and a binder production adds one more. Other occurrences of M receive the same body with metavariable-free arguments, which DT° allows. So $\tau_r \in \text{DT}^\circ$. *Instances.* $\text{inst}(\tau) = \bigsqcup_r \text{inst}(\tau_r)$: at a pattern occurrence of M an instance shows the root production of $\theta(M)$, and distinct productions give distinct subtrees there (for a hole production z_i , the distinct bound variable y_i). *Probabilities.* For $s \in \text{inst}(\tau_r)$ with τ 's matcher θ , the matcher of τ_r gives the children's bodies of $\theta(M)$, and the root factorisation gives $\mathcal{Q}(\theta) = p_r \mathcal{Q}_{\tau_r}(s)$. (Holes and bound indices are drawn alike, so a child body under a binder has the law of a body with one more argument.) Hence $w_\tau \mathcal{Q}_\tau(s) = \sum_r w_\tau p_r \mathcal{Q}_{\tau_r}(s)$, and the other components are unchanged. *Under L1, L2.* A theory-citation node's conclusion has the same law under (T, w) and (T', w') , by the L0 statement; every other node has the same law, and a tree's validity and conclusion depend only on its nodes' conclusions. So $\mu_{T'} = \mu_T$ and $Z_{T'} = Z_T$. (c) A renaming permutes the factors of $\mathcal{Q}(\theta)$; an argument permutation permutes the holes, which a hole-symmetric $\mathcal{Q}$ ignores. The converse is (a) for single templates; for first-order templates with only 0-ary term metavariables, AS Prop C.1 (under its richness condition) turns equal instance sets into $\tau \equiv \tau'$. That proposition assumes one side has only 0-ary term metavariables (referee m10), so it says nothing about templates such as $A1 = B \rightarrow (C \rightarrow B)$. $\square$

With *learned* weights the likelihoods differ; that case is Propositions 1.14 and 1.15. The pre-referee sentence “the likelihood never tells a schema from its split” holds only at matched fixed weights (referee m14). Track pa's Prop 2.1 is (b) for T_{Ind} with a shared grammar $\mathcal{Q}(\varphi) = \mathcal{Q}(f)\mathcal{Q}(\cdot \mid f)$: an induction datum with motive root f has probability $w\mathcal{Q}(f)\mathcal{Q}(\varphi \mid f) = w\mathcal{Q}(\varphi)$ under both.

Computed (c2, Part A): $z+0 = z$ against its split over $0, S, +, \cdot$ with root law $(0.5, 0.3, 0.1, 0.1)$; maximal difference $1.08 \cdot 10^{-19}$ over all 2849 instances with $|t| \leq 9$, every instance matched against every template.

Proof of Remark 1.8 (experiments Prop X13). Under $\mathcal{Q}$ a closed term has root f with probability q_f and, given the root, children drawn independently from the same closed-term law (every node of a closed term has the same context). So “cite τ with probability w_τ and draw the body of u from $\mathcal{Q}$ ” and “cite τ_f with probability $w_\tau q_f$ and draw the bodies of $u_1, \dots, u_r$ from $\mathcal{Q}$ ” give the same law on the cited sentence, jointly with the bodies of τ 's other metavariables. Every later chain step depends only on the current sentence and on the set of MP components, which is the same in T and T' because neither τ nor any τ_f is an MP component. $\square$

A.3 Identification in the limit and Gold

Proof of Theorem 1.9(a). By Theorem 1.2, $\pi_n(C^*) \rightarrow 1$, and every $T \in C^*$ has $\text{supp } P_T = \text{supp } P_{T^*}$. At most one L can have posterior mass above $\frac{1}{2}$; once $\pi_n(C^*) > \frac{1}{2}$ it is $\text{supp } P_{T^*}$, which is $\bigcup \text{inst}(T^*)$ under L0 (??(c)) and $\text{Th}(T^*)$ under L1 (??(c)). $\square$

Proof of Theorem 1.9(b). The proof of Theorem 1.2, with the frequency functional replaced by a support functional. *A countable parameter that the data determine.* Put $U(T, w) := \bigcup \text{inst}(T)$; it takes countably many values. For a sequence x let $F'(x) := \{x_i : i \geq 1\}$. For w in the open simplex, $\text{supp } P_{T,w} = \bigcup \text{inst}(T)$ is a countable set of points of positive probability, each of which occurs a.s., and nothing else occurs; so $P_{T,w}^\infty(F' = U(T, w)) = 1$. Dir_T charges only the open simplex. Hence $\Pi(U = F'(X)) = 1$ for the joint law Π of (T, w, X) . *Steps 1 and 3.* The posterior given D_n is $\Pi(\cdot \mid D_n)$ (Bayes' formula with the likelihood as a density with respect to the prior). For each of the countably many sets $A_\nu := \{(T, w) : \bigcup \text{inst}(T) = \nu\}$, Lévy's theorem gives $\pi_n(A_\nu) \rightarrow \Pi(A_\nu \mid X_\infty)$ M -a.s. Since $F'(X)$ is X_∞ -measurable, $1 = \mathbb{E}[1\{U = F'(X)\} \mid X_\infty] = \sum_\nu 1\{F'(X) = \nu\} \Pi(A_\nu \mid X_\infty)$ a.s. So, M -a.s., $\pi_n(A_{F'(X)}) \rightarrow 1$. *Step 4, changed.* Let E be the M -null event where this fails. Since $M \geq \pi(T^*) \int P_{T^*,w}^\infty(\cdot) \text{Dir}_{T^*}(dw)$, we get $\int P_{T^*,w}^\infty(E) \text{Dir}_{T^*}(dw) = 0$, so $P_{T^*,w}^\infty(E) = 0$ for Dir_{T^*} -a.e. w , which is Lebesgue-a.e. w because the Dirichlet density is positive on the open simplex. For such w^* , a.s. $F'(X) = \bigcup \text{inst}(T^*)$ and $\pi_n(A_{\bigcup \text{inst}(T^*)}) \rightarrow 1$. $\square$

The pre-referee proof said that the steps of Theorem 1.2 “go through with the countable sum replaced by an integral”. That sentence is withdrawn: Step 3 needs a countable family of sets, which the support functional provides and the law itself does not (referee M4, who suggested this route).

Sketch of Theorem 1.9(b'). $P_{T,w}$ lies in the Polish space of probability vectors on $\mathcal{S}$ (ℓ^1 metric), and the frequency functional F recovers it a.s. For a countable convergence-determining family of bounded continuous functions g , Lévy's theorem gives that the posterior mean of $g(P)$ converges to $g(F(X))$ a.s. So the posterior law of $P_{T,w}$ converges weakly to the point mass at P^* , which gives the statement for total-variation balls; transfer to Lebesgue-a.e. w^* as in (b). Not written out: the choice of the family and the passage from weak convergence to total-variation balls.

Proof of the counter-statement in Remark 1.10. Fix T . The set $W_T := \{w : P_{T,w} = P^*\}$ is the intersection of the simplex with an affine subspace, since $w \mapsto P_{T,w}$ is affine. If it had positive Lebesgue measure in the simplex, the affine map would be constant on an open set, hence on the whole simplex, and every component of T would have law P^* (for $|T| = 1$ the condition is $\mathcal{Q}_\tau = P^*$). The posterior on w given T is absolutely continuous with respect to Dir_T . So the posterior mass of the exact class is 0 at every n unless some template σ has $\mathcal{Q}_\sigma = P^* = \sum_\tau w_\tau^* \mathcal{Q}_\tau$. For each σ , linear independence of $\{\mathcal{Q}_\tau : \tau \in T^*\}$ allows at most one such w^* , and there are countably many σ . $\square$

Proof of Proposition 1.12. (a) is Theorem 1.2. (b) Enumerate $L_\infty = \{e_1, e_2, \dots\}$ and build the text in stages. At stage i , with σ the prefix built so far, pick $k_i > k_{i-1}$ with $\sigma \cup \{e_i\} \subseteq L_{k_i}$ (possible since the chain increases to L_∞) and append e_i ; the new prefix σ' has positive P_{k_i} -probability. By (a), applied to $P_{k_i}^\infty$ conditioned on the positive-probability prefix σ' , almost every continuation reaches $g_n = L_{k_i}$ at some finite n ; pick such a continuation, with elements in $L_{k_i} \subseteq L_\infty$, and cut it at a time where $g_n = L_{k_i}$. The resulting sequence lists every e_i and only elements of L_∞ , and $g_n = L_{k_i} \neq L_\infty$ at infinitely many times. (c) By (a) with $i = \infty$, P_∞^∞ -a.s. $g_n = L_\infty$ eventually. $\square$

Computed (E5; ??): on data uniform on L_5 the mass of L_5 is $2 \cdot 10^{-6}$ at $n = 64$, 0.661 at 128 and 1.000 from 256 in all 10 seeds (L_5 gains $3 - \log_2 5 \approx 0.68$ bits per datum, repaying 72 prior bits and its Dirichlet cost); on Gold's text against the MAP learner over $\{L_1, \dots, L_{60}, L_\infty\}$

the first 14 stages end (740 data), which shows only that these stages end (experiments' referee m10). *The $\text{I}\Sigma_n$ chain* (model §5.2, heuristic). Under L1 with theorem data, PA's generator charges $\text{Th}(\text{PA}) \setminus \text{Th}(\text{I}\Sigma_n)$, so $\text{I}\Sigma_n$ is eventually rejected on PA-data; and PA's generator charges $\text{Th}(\text{PA}) \setminus \text{Th}(\text{I}\Sigma_5)$, so PA is eventually rejected on $\text{I}\Sigma_5$ -data. Both masses presumably are tiny, as they need long derivations with complex induction; if so, the number of data needed is of order $1/P_{\text{PA}}(\text{Th}(\text{PA}) \setminus \text{Th}(\text{I}\Sigma_n))$. The masses were not estimated.

A.4 Splits

Proof of Proposition 1.14. (a) The split's instance sets are disjoint (Proposition 1.7(b)), so the labelling sum of ?? has one term, and $\mathcal{Q}_{\tau_f}(s) = \mathcal{Q}_{\tau}(s)/p_f$ for $s \in \text{inst}(\tau_f)$. So $P_{T'}^{\text{Dir}}(D) = \text{DirMult}(n_f; \alpha) \prod_i \mathcal{Q}_{\tau}(X_i)/p_{f_i}$, while $P_T(D) = \prod_i \mathcal{Q}_{\tau}(X_i)$. (b) Stirling: $\ln \Gamma(a+m) = (a+m - \frac{1}{2}) \ln m - m + \frac{1}{2} \ln 2\pi + o(1)$ as $m \rightarrow \infty$. Applied to $\ln \Gamma(K\alpha + n)$ and the K terms $\ln \Gamma(\alpha + n_f)$,

$$\begin{aligned} \ln \text{DirMult}(n_f; \alpha) &= \sum_f n_f \ln n_f - n \ln n + (\alpha - \frac{1}{2}) \sum_f \ln n_f \\ &\quad - (K\alpha - \frac{1}{2}) \ln n + \frac{K-1}{2} \ln 2\pi + \ln \frac{\Gamma(K\alpha)}{\Gamma(\alpha)^K} + o(1). \end{aligned}$$

With $\ln n_f = \ln n + \ln \hat{r}_f$ the first two terms are $n \sum_f \hat{r}_f \ln \hat{r}_f$, and the $\ln n$ terms collect to $(K\alpha - \frac{K}{2} - K\alpha + \frac{1}{2}) \ln n = -\frac{K-1}{2} \ln n$. Subtracting $\sum_f n_f \ln p_f$ turns $n \sum_f \hat{r}_f \ln \hat{r}_f$ into $n \text{KL}(\hat{r}||p)$. (c) Pearson's and Wilks's χ^2 limit (known), with all $n_f \rightarrow \infty$ a.s. since p is in the open simplex. (d) The strong law for $\hat{r}$; $\ln \text{DirMult}(n_f; \alpha) = n \sum_f \hat{r}_f \ln \hat{r}_f + O(\ln n)$ also when some $r_f = 0$. (e) The prior. $\square$

With $\alpha = \frac{1}{2}$ the term $(\alpha - \frac{1}{2}) \sum \ln \hat{r}_f$ vanishes, and in the well-specified case $\mathbb{E}\Delta_n \approx \frac{K-1}{2} - \frac{K-1}{2} \ln n + c_{K,\alpha}$; for $K = 4$, $c_{4,1/2} = \ln \pi^{-2} + \frac{3}{2} \ln 2\pi = 0.467$, which gives the predictions of Table 2.

n	10^2	10^3	10^4	10^5	10^6
well specified, mean Δ_n	-4.93	-8.34	-11.77	-15.27	-18.85
prediction	-4.94	-8.39	-11.85	-15.30	-18.76
$P(\Delta_n > 0)$	0.007	0	0	0	0
misspecified, mean Δ_n	-0.06	39.7	482.2	4917.5	49356.3
prediction	-1.50	39.5	480.4	4920.4	49351.8

Table 2: Proposition 1.14, computed (model check c2, Part B): Δ_n in nats, $K = 4$, $\alpha = \frac{1}{2}$, $\mathcal{Q}$'s root law $p = (0.5, 0.3, 0.1, 0.1)$, 400 runs per n . Well specified: $r = p$. Misspecified: $r = (0.4, 0.3, 0.2, 0.1)$, $\text{KL}(r||p) = 0.0494$ nats; predictions from (b) with $\hat{r}$ replaced by r , plus $\frac{K-1}{2}$ (the mean of $\chi_{K-1}^2/2$) in the well-specified case; at $n = 100$ the misspecified expansion is not yet accurate. The referee (r4) checked (b) for $\alpha = 0.3, 1, 2$ ($K = 3$): the residual tends to 0 like $1/n$.

Proof of Proposition 1.15. (a) Under (T', w') a theory-citation node chooses τ_r with probability w'_r and draws the children's bodies from $\mathcal{Q}$. By the root factorisation and $\text{inst}(\tau) = \bigsqcup_r \text{inst}(\tau_r)$, its conclusion has the law of a citation of τ whose M has root law w' . All other nodes have the same laws, and validity and conclusions depend only on the nodes' conclusions; so μ and Z coincide, under L0, L1 and L2. (b) By Tonelli,

$$\mathbb{E}[\text{BF}_{n+1} \mid D_n] = \int \prod_{i \leq n} \frac{P(p')}{P(p)}(X_i) \sum_{x: P(p)(x) > 0} P(p')(x) \text{Dir}(dp') \leq \text{BF}_n.$$

Ville’s inequality (known; IL Lemma C.4) and the supermartingale convergence theorem (known).
(c) Write

$$\ln \text{BF}_n = \ln \frac{P_{T'}^{\text{Dir}}(D_n)}{P^{(r)}(D_n)} + \sum_{i \leq n} \ln \frac{P^{(r)}(X_i)}{P^{(p)}(X_i)}.$$

The second sum divided by n tends to $\text{KL}(P^{(r)} \| P^{(p)})$ a.s., as in Proposition 1.5(a). For the first term: *upper bound*: $P_{T'}^{\text{Dir}}(D_n)/P^{(r)}(D_n)$ is a nonnegative supermartingale under $P^{(r)}$, as in (b); it converges a.s. to a finite limit, so $\limsup_n n^{-1} \ln$ of it is ≤ 0 . *Lower bound*: for $\gamma > 0$ let $U_\gamma := \{p' : \text{KL}(P^{(r)} \| P^{(p')}) < \gamma\}$. By the continuity lemma below it contains a neighbourhood of the interior point r , so $\text{Dir}(U_\gamma) > 0$; let Dir_U be the normalised restriction. By Jensen’s inequality,

$$\ln \int_{U_\gamma} \prod_i \frac{P^{(p')}(X_i)}{P^{(r)}(X_i)} \text{Dir}(dp') \geq \ln \text{Dir}(U_\gamma) + \sum_i Y_i, \quad Y_i := \int \ln \frac{P^{(p')}(X_i)}{P^{(r)}(X_i)} \text{Dir}_U(dp').$$

The Y_i are i.i.d., and by Tonelli on the positive and negative parts

$$\mathbb{E}Y^+ \leq \int \mathbb{E} \frac{P^{(p')}(X)}{P^{(r)}(X)} \text{Dir}_U(dp') \leq 1, \quad \mathbb{E}Y = - \int \text{KL}(P^{(r)} \| P^{(p')}) \text{Dir}_U(dp') \geq -\gamma.$$

By the strong law, $\liminf_n n^{-1} \ln[P_{T'}^{\text{Dir}}(D_n)/P^{(r)}(D_n)] \geq -\gamma$ a.s.; let $\gamma \downarrow 0$ along a sequence. *Continuity lemma*: $\text{KL}(P^{(r)} \| P^{(p')}) \rightarrow 0$ as $p' \rightarrow r$. For a tree π , $\text{Pr}_{p'}(\pi)/\text{Pr}_r(\pi) = \prod p'_f/r_f$ over π ’s theory-citation leaves, f the root production chosen there. Let $L(\pi)$ be the number of these leaves, $\varrho := \min_f p'_f/r_f \leq 1$ and $\varrho' := \max_f p'_f/r_f \geq 1$. Then $\mu_{p'}(x) \geq \sum_{\pi \rightarrow x} \varrho^{L(\pi)} \text{Pr}_r(\pi) \geq \mu_r(x) \varrho^{\mathbb{E}_r[L|x]}$ (Jensen, $\mathbb{E}_r[\cdot | x]$ over the valid trees concluding x), and $Z_{p'} \leq \mathbb{E}_r[\varrho'^L; \text{valid}]$. So

$$\text{KL}(P^{(r)} \| P^{(p')}) \leq \ln \frac{1}{\varrho} \cdot \mathbb{E}_r[L | \text{valid}] + \ln \frac{\mathbb{E}_r[\varrho'^L; \text{valid}]}{Z_r}.$$

Here $\mathbb{E}_r[L | \text{valid}] \leq \mathbb{E}_r[N]/Z_r < \infty$, with N the number of derivation nodes ($m < 1$). And $\mathbb{E}_r[\varrho'^L; \text{valid}] \rightarrow Z_r$ as $\varrho' \downarrow 1$ by dominated convergence, since $L \leq N$ and $\mathbb{E}_r[e^{\zeta N}] < \infty$ for small $\zeta > 0$ (??(d) for the derivation tree). Both terms tend to 0 as $p' \rightarrow r$. Under L2, $L \leq 2^d$ and the lemma is immediate. $\square$

The pre-referee notes marked “a derivation-based likelihood does not change the MDL finding” as proved on the strength of the matched-weight tie alone; that was withdrawn in favour of this proposition (referee M3), and (d) is a conjecture.

part	result
(a)	split with weights $(u, 1 - u)$ against the schema with root law u , $u \in \{0.1, 0.37, 0.8\}$: maximal difference $1.05 \cdot 10^{-14}$
latency	88 of 5885 conclusions depend on u through derived trees
(b)	$\delta' = 0.5, 0.2, 0.05$: $P(\max_{n \leq 3000} \text{BF}_n \geq 1/\delta') = 0.427, 0.133, 0.036$ (1000 runs, every n checked)
(c)	data root law 0.3, schema 0.6: $\text{KL} = 0.11487$ nats; mean $n^{-1} \ln \text{BF}_n = 0.11301, 0.11460, 0.11464$ at $n = 10^3, 10^4, 10^5$
(d)	mean $\ln \text{BF}_n = -1.765, -6.405$ at $n = 10^2, 10^6$ (400 runs); slope -0.507 (-0.498 on $n \geq 10^4$) against -0.5

Table 3: Proposition 1.15, computed (model check c13): a propositional L2 model, depth 2, atom a , logical axioms A1–A3 instantiated from a grammar truncated to size ≤ 5 , MP; $\tau = F \rightarrow F$ with F rooted at a or $\neg$, $K = 2$, root law $p_a = 0.6$; 5885 possible conclusions. Citations are latent only weakly here.

E8 windows. In Remark 1.16 the two windows ($n = 256 \dots 4096$: 1.43, 1.51 bits per doubling; 1024 $\dots$ 4096: 1.60, 1.76) are both correct for their windows; the prior, not included, adds 76.4 bits for whole. Proposition 1.15(b) is a time-uniform bound for a one-component schema with fixed weights, a different setting, so E8 is not a test of it.

A.5 Spare templates

Proof of Proposition 1.18(a). The labellings of the data by templates of $T^* \cup \{\sigma\}$ are those for T^* , with $n_\sigma = 0$. For each labelling, the Dirichlet moment of ?? for $T^* \cup \{\sigma\}$ divided by that for T^* is $\frac{\Gamma(A+\alpha_\sigma)}{\Gamma(A+\alpha_\sigma+n)} \frac{\Gamma(\alpha_\sigma)}{\Gamma(\alpha_\sigma)} \bigg/ \frac{\Gamma(A)}{\Gamma(A+n)}$, the same for every labelling; factor it out. Asymptotics: $\Gamma(a+n)/\Gamma(b+n) \sim n^{a-b}$. $\square$

For a symmetric prior ($\alpha_\tau = \alpha_\sigma = \alpha$, $A = K\alpha$) this is experiments' Prop X7(a) and, at $\alpha = \frac{1}{2}$, pa's Prop 5.1: $\log_2$ of the inverse is $\frac{1}{2} \log_2 n + \log_2 \frac{\Gamma(K/2)}{\Gamma((K+1)/2)} + o(1)$ bits. In pa's code, with $K = 8$ and the spare $F \rightarrow F$ (15 prior bits), this predicts 19.03 and 19.82 bits at $n = 1000$ and 3000; pa computed 19.0 and 19.8.

Proof of Proposition 1.18(d1). $P_{T^* \cup \{\sigma\}, w} = \sum_{\tau \neq \tau_1} w_\tau \mathcal{Q}_\tau + (w_{\tau_1} + w_\sigma) \mathcal{Q}_{\tau_1}$ for every w . By the aggregation property of Dirichlet laws, $(w_{\tau_1} + w_\sigma, (w_\tau)_{\tau \neq \tau_1}) \sim \text{Dir}(\alpha_1 + \alpha_\sigma, (\alpha_\tau)_{\tau \neq \tau_1})$. So $P_{T^* \cup \{\sigma\}}^{\text{Dir}}$ is the Dirichlet marginal of T^* with α_1 replaced by $\alpha_1 + \alpha_\sigma$, and with disjoint instance sets the one-labelling formula of ?? gives, for every D_n with n_1 data in $\text{inst}(\tau_1)$, exactly

$$R_n = \frac{\Gamma(A + \alpha_\sigma) \Gamma(A + n)}{\Gamma(A) \Gamma(A + \alpha_\sigma + n)} \cdot \frac{\Gamma(\alpha_1 + \alpha_\sigma + n_1) \Gamma(\alpha_1)}{\Gamma(\alpha_1 + \alpha_\sigma) \Gamma(\alpha_1 + n_1)}.$$

The limit follows from $\Gamma(a+n)/\Gamma(b+n) \sim n^{a-b}$ and $n_1/n \rightarrow w_1^*$ a.s. $\square$

We checked the limits in Table 4 against this formula: with $A = 1$, $\alpha_1 = \frac{1}{2}$, $w_1^* = \frac{1}{2}$ it gives $\ln R_\infty = \ln(\frac{\pi}{2} \cdot 2^{-1/2}) = 0.1050$ at $\alpha_\sigma = \frac{1}{2}$, 0 at $\alpha_\sigma = 1$ and $\ln \frac{2}{3} = -0.4055$ at $\alpha_\sigma = 2$, as c15 reports.

Sketch of Proposition 1.18(b), (c). Write $w = ((1-u)\tilde{w}, u)$ with $u \sim \text{Beta}(\alpha_\sigma, A)$ independent of $\tilde{w} \sim \text{Dir}(\alpha_\tau)$ (aggregation). The per-datum factor is $1 - u + u \mathcal{Q}_\sigma(X)/P_{T^*, \tilde{w}}(X)$. In (b), at $\tilde{w} = w^*$ its mean is $1 - uc$, so the likelihood decays like e^{-ncu} in u , and $\int u^{\alpha_\sigma-1} e^{-ncu} du \sim \Gamma(\alpha_\sigma)(cn)^{-\alpha_\sigma}$. In (c) the mean is 1 (zero drift) and the log-likelihood in u is locally $-Iu^2/2 + \sqrt{n}Zu$; the integral against $u^{\alpha_\sigma-1}$ scales as $n^{-\alpha_\sigma/2}$. Not written out: uniformity in $\tilde{w}$ near w^* , and almost-sure or in-probability control of the fluctuations. This is a boundary Laplace approximation of the kind analysed for overfitted mixtures by Rousseau and Mengersen (2011), whose setting has continuous component parameters (checked by the model track's referee); their result is not applied here.

Sketch of Proposition 1.18(d2). T^* and $T^* \cup \{\sigma\}$ parametrise the same family of laws, so the two marginals integrate the same likelihood against two prior densities on that family, and Laplace's method gives the ratio of the densities at P^* . Not written out: the Laplace step.

A.6 Misspecification

Proof of Theorem 1.20. On $\text{supp } P_H$, $\ln(P_T/P_{T_0}) = Y_0 - Y_T$ with $Y_T := \ln(P_H/P_T)$. Y_T has an integrable negative part, $\mathbb{E}(\ln \frac{P_T}{P_H})^+ \leq \mathbb{E} \frac{P_T}{P_H} \leq 1$, and mean K_T ; $Y_0 \in L^1$ because also

case	source	result
(a)	E5(a)	$0 = S0$ never cited, $u \cdot 0 = 0$ never matched: $\log_2$ BF $-1.00, -6.83$ at $n = 1, 4096$, equal to the formula on every seed; slope -0.500
(a), (b)	c3, r4	formula matched to 4 decimals; slopes $-0.499, -0.498$; referee: $-\alpha_\sigma$ to 3 decimals for $\alpha_\sigma = \frac{1}{2}, 1, 2$
(c)	E5(a2)	nested $\varphi(Sz)$ in $\{\varphi(z)\}$: mean $\log_2$ BF -2.196 at $n = 10^2, -7.200$ at 10^8 (1000 draws per n); slope -0.251 ± 0.002
(c)	c3b, r4	-0.265 on $10^3 \dots 10^7$ (s.e. ≈ 0.07 per point); referee $-0.230, -0.523, -0.961$ for $\alpha_\sigma = \frac{1}{2}, 1, 2$
(d1)	c15	formula equals the numerical integral to 5 decimals; at $n = 10^6$ within 0.003 of the limit
(d2)	c15	$w_1^* = 0.3$: $\ln R_n$ changes by $-0.031, -0.021, -0.004$ per decade; $w_1^* = \frac{1}{2}$ ($P^* = \mathcal{Q}_\sigma$, density infinite): $+0.22, +0.20, +0.13$

Table 4: Spare templates, computed. Slopes are of the mean log Bayes factor against $\log n$ and so do not depend on the base. The primary estimate for (c) is E5(a2); its rate remains a proof sketch.

$K_{T_0} < \infty$. By the strong law, $n^{-1} \sum_{i \leq n} (Y_{0,i} - Y_{T,i}) \rightarrow K_{T_0} - K_T$ a.s., the value $-\infty$ allowed when $K_T = \infty$. $\mathcal{H}$ is finite. $\square$

Inside $\mathbb{M}$, members with equal KL but different laws trade places like a zero-drift random walk.

Proof of Example 1.21. $P_H = P_\sigma$, so Theorem 1.2 applies with truth σ . Every $T \in C_\sigma$ has $\bigcup \text{inst}(T) = \text{inst}(\sigma) = I_c(0 + x = x)$ by Proposition 1.7(a) (parameters not admissible), hence $\text{Th}(T) = \text{Th}(I_c(0 + x = x))$. Take the structure $\mathbb{N} \cup \{a\}$, standard on $\mathbb{N}$, with $0 + a := 0$ and the other operations extended arbitrarily, for instance $Sa := a$ and $a \circ y := y \circ a := a$ for $\circ \in \{+, \cdot\}$ except $0 + a = 0$. Closed terms denote standard numbers, so every closed instance holds; $\forall x(0 + x = x)$ fails at a . By Corollary 1.4 the posterior probability that $T \vdash \forall x(0 + x = x)$ tends to 0. $\square$

Proof of Lemma 1.22. Let τ cover some $t = 0 \in Z$. *Shape.* The datum has no binders, so τ 's rigid skeleton has none. A pattern occurrence of positive arity needs bound variables in scope, so every metavariable is 0-ary. The only formula position in $t = 0$ is the root. *Root metavariable.* Then τ is a formula metavariable, with the false instance $0 = S0$. *Otherwise* $\tau = (u = v)$ with term templates u, v . u matches t , so its rigid nodes are $+$ and 0 and its metavariable occurrences are leaves; hence $\text{val}(u\theta) = \sum_j c_j \text{val}(\theta(z_j))$, c_j the multiplicity of z_j in u . v matches 0 , so v is 0 or a metavariable z . $v = 0$. Soundness requires $\sum_j c_j \text{val}(\theta(z_j)) = 0$ for all θ ; $\theta(z_j) := S0$ for all j shows $c_j = 0$. So τ is ground and covers one sentence. $v = z$, z not in u . The value of $u\theta$ does not depend on $\theta(z)$; one of $\theta(z) \in \{0, S0\}$ gives a false instance. $v = z$, z in u . Soundness requires $\sum_j c_j \text{val}(\theta(z_j)) = \text{val}(\theta(z))$ for all θ ; unit test values give $c_z = 1$ and $c_j = 0$ for $j \neq z$. So u contains z once and is otherwise ground; covering $t = 0$ forces $\theta(z) = 0$, and τ covers exactly one member of Z . $\square$

Proof of Example 1.23. By Lemma 1.22 each T_j covers at most $|T_j|$ members of Z . Since $\text{supp } P_H$ is infinite, $P_H(Z \setminus \bigcup_j \text{inst}(T_j)) > 0$, so a.s. some datum falls outside every T_j , which then has likelihood 0 under $\mathbb{L}_0$. $P_{T_{\text{esc}}} > 0$ on Z (full-support $\mathcal{Q}$, $z := t$). So $\pi_n(T_{\text{esc}}) = 1$ from

that datum on. $T_{\text{esc}} \vdash S0 = 0$ (instance $z := S0$); $Q \vdash \neg S0 = 0$ (Q1). Q 's axioms are sentences outside Z , so a theory containing only them has likelihood 0 at the first datum. $\square$

Computed (c4, Part 2): twenty sound theories (ground equations for the 20 most likely data, which carry mass 0.956, weighted as the human's conditional law) share prior 0.99 against T_{esc} ; $S0 = 0$ reached posterior ≥ 0.99 in 2000/2000 runs, median 17 data, 90.2% by $n = 50$. ?? uses this example for the verifier.

seed	$\lambda = 1$			$\lambda = 2$		
	11	12	13	11	12	13
distinct data at $n = 10^4$	282	325	330	282	325	330
hybrid – memorisation, $n = 10^3$	≥ 1143	≥ 1207	≥ 1065	≥ 3344	≥ 3505	≥ 3063
hybrid – memorisation, $n = 10^4$	≥ 7370	≥ 8613	≥ 8625	≥ 21703	≥ 24719	≥ 24596
escape alone – memorisation, $n = 10^4$	-4234	-2649	-2719	12871	16289	16120

Table 5: Remark 1.24, computed (model check c7): $\log_2$ score (prior plus L0 marginal likelihood, $\text{Dir}(\frac{1}{2})$) relative to memorising all distinct data. Human data $t = 0$, t a $\{0, +\}$ -tree (0 with probability 0.7); model Q root law (0.5, 0.3, 0.1, 0.1) on $(0, S, +, \cdot)$; prior $2^{-\lambda \ell}$ under a simplified code (3 bits per symbol, 2 per metavariable occurrence, Elias γ for the number of templates), not π_λ of ?? . Hybrid: ground templates for data seen at least twice plus $z = 0$; its score is a lower bound (one labelling). Escape alone: $\{z = 0\}$. Positive favours the unsound theory. At $\lambda = 1$ the escape alone loses to memorisation by $n = 10^4$, but the hybrid does not.

Details of Example 1.25. We write $T_{Q0} := T_Q \cup \{0 + z = z\}$ and $T_{\text{mix}} := T_Q \cup \{z = 0\}$. Each node of the $L1^{\text{eq}}$ grammar cites a template of T (probability α_{ax} , metavariables from Q), or applies refl ($t \sim Q$), sym, trans (valid iff the middle terms agree), congruence for S , or congruence for $+$ (a side term $u \sim Q$, left or right). A tree is valid iff its steps are valid and every term lies in U_N , the closed terms of size $\leq N$. $P_T = \mu_T / Z_T$, with μ_T the least fixpoint of the grammar's equation. These rules are complete for closed equational consequence (Birkhoff; recalled, not checked against the source (Birkhoff, 1935)). A Monte Carlo simulation agrees with the fixpoint ($Z = 0.54909$ against 0.54965 ± 0.00091 ; $|z| \leq 1.31$ on eight conclusion frequencies). Human law: $t = 0$, t a closed $\{0, +\}$ -term from a grammar choosing 0 with probability $1 - h$ and $+$ with h , conditioned on size $\leq N$, $h \in \{0.3, 0.45\}$, with and without the datum $0 = 0$.

A.7 Predictive versus deductive

Proof of Theorem 1.28. (a) is Proposition 1.5(b) with the index shifted; (b) is Corollary 1.4 with $d = \infty$, and the two cases are those named after it. (c) $\mathbb{E}_{P_{T_1}} \ln \frac{P_{T_1}(X)}{P_{T_2}(X)} = \text{KL}(P_{T_1} \| P_{T_2})$ per datum, summed over the data; Pinsker's inequality $\text{TV} \leq (\text{KL}/2)^{1/2}$ (known). For the misspecified remark: $M(D_n) \geq \pi(T)P_T(D_n)$ for every T and every sequence, so the cumulative log loss of M exceeds that of T by at most $\ln(1/\pi(T))$. $\square$

Proof of Proposition 1.29. (a) A tree's probability is the product of its nodes' probabilities. A theory-citation leaf whose template is not recorded contributes $\alpha_{\text{ax}} \sum_\tau w_\tau Q_\tau(\text{concl}(v)) = \alpha_{\text{ax}} P_T^{\text{L0}}(\text{concl}(v))$, the sum over the unrecorded choices factorising over leaves (matching is unique, so $Q_\tau(\text{concl}(v))$ is the sum over substitutions). No other node depends on T . Validity depends only on conclusions, so $Z_T = \Pr[\text{valid}]$ is a function of the law of theory-leaf conclusions, i.e. of

grammar	rule probabilities (ax, refl, sym, trans, cong _S , cong ₊)	$\mathcal{Q}$ root law (0, S , +)	minimiser	settings
1	(.35, .15, .1, .15, .1, .15), $m = 0.65$	(.55, .2, .25)	T_{esc} with $0 = 0$; T_{mix} without	8/8 unsound
2	(.5, .1, .05, .2, .05, .1), $m = 0.60$	(.55, .2, .25)	as grammar 1	8/8 unsound
3	as grammar 1	(.8995, .1, .0005)	T_{Q0} in 5; T_{mix} in 3 (all with $0 = 0$)	5/8 sound

	mean $\ln P_T(t = 0)$ by size of t	1	3	5	7	9
grammar 1: T_Q		-1.88	-1.82	-8.45	-15.52	-22.89
grammar 1: T_{esc}		-0.80	-3.23	-5.22	-7.20	-9.19
grammar 3: T_Q		-1.62	-1.57	-7.91	-15.41	-23.33
grammar 3: T_{Q0}		-1.63	-1.28	-7.34	-14.55	-22.19
grammar 3: T_{esc}		-0.43	-8.58	-16.30	-24.01	-31.72

Table 6: Example 1.25, computed (model check c16). Top: the cross-entropy minimiser over $\{T_Q, T_{Q0}, T_{\text{esc}}, T_{\text{mix}}\}$ in the eight settings of each grammar ($N \in \{7, 9\}$, $h \in \{0.3, 0.45\}$, with and without $0 = 0$). Bottom: per-datum log-likelihoods at $N = 9$. In grammar 1, T_Q pays about 7 nats per extra + and T_{esc} about 2; e.g. at $N = 9$, $h = 0.45$, data $t \neq 0$, the cross-entropies are 7.758 (T_Q), 7.124 (T_{Q0}), 4.951 (T_{esc}), 4.904 (T_{mix}) nats, so the odds of T_Q against T_{mix} fall by about 2.85 nats per datum. In grammar 3, T_{esc} pays $\ln \frac{1}{p+p_0} \approx 7.7$ nats per + through $\mathcal{Q}$; T_Q 's increments (6.3, 7.5, 7.9 nats between consecutive sizes) are still rising at size 9, so the large-term limit in grammar 3 is not settled.

$P_T^{\text{L}0}$. (b) “If” is (a). “Only if”: the one-node tree citing s as a theory axiom has probability $\alpha_{\text{ax}} P_T^{\text{L}0}(s)/Z_T$; equality for all s makes $P_T^{\text{L}0}$ and $P_T^{\text{L}0}$ proportional, and both sum to 1. The rest is Theorem 1.2, Corollary 1.3 and Proposition 1.7(a), (b). (c) Proposition 1.6. $\square$

If the template name is recorded at each citation leaf, the argument is easier.

Proof of Remark 1.30. P_T^K is a probability on $\mathcal{S}$ whose normaliser does not depend on the data. The proofs of Theorem 1.2, Corollary 1.4, ?? and ?? use only that each hypothesis carries a fixed law on a countable set and that the data are i.i.d. from one of them; they therefore hold with P_T^K in place of P_T when the data are i.i.d. from $P_{T^*}^K$. $\square$

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